



Assignment

Course Module: HDIT 21132

Emerging Technology

Higher Diploma in Information Technology

Batch -01, 2025

Department of Information Technology

Faculty of Computing

International Campus of Science and Technology

This report submitted as a partial fulfilment of the requirements
for ICST for Higher Diploma in Information Technology.

I certify that the content of this report is all my own work and
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prescribed manner.

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Module Name: Emerging Technology

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Chapter 1 – Introduction

Problem Background:

Traditional homes require manual control of appliances, resulting in energy waste and reduced security. Emerging technologies such as IoT, VR and AR provide intelligent automation and improved user interaction.

Problem Statement:

Homeowners need a system that can monitor environmental conditions, automate devices and provide an interactive way to understand and control the smart home.

Objectives:

- Design an IoT-based smart home.
- Create a VR visualization.
- Develop an AR information overlay.
- Demonstrate integration of all three technologies.

Scope:

The project includes a home gateway, sensors, smart devices, VR visualization and AR interaction.

Chapter 2 – Proposed Solution

System Overview:

The proposed Smart Home integrates Cisco Packet Tracer (IoT), Delightex EDU (VR) and MyWebAR (AR). Motion, temperature and smoke sensors collect data. Smart devices respond automatically according to predefined rules.

Technologies Used:

- Cisco Packet Tracer
- Delightex EDU
- MyWebAR

System Architecture:

Sensors -> Home Gateway -> Smart Devices -> User Dashboard -> VR Visualization -> AR Information Overlay.

Chapter 3 – IoT Simulation

Design:

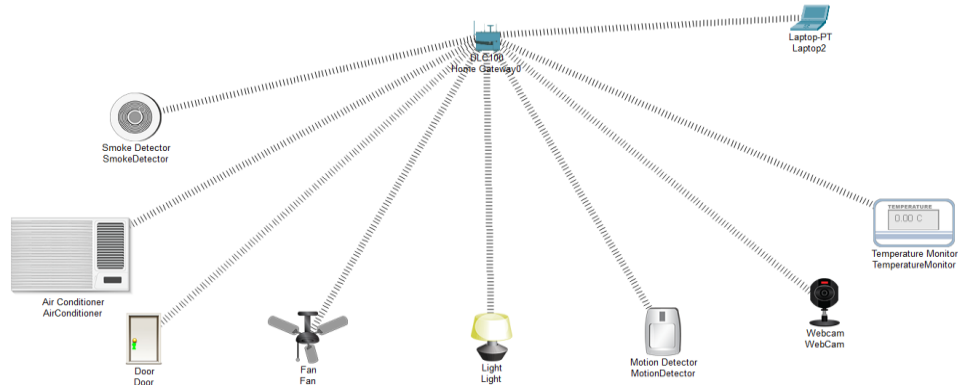
The smart home contains a Home Gateway, Laptop, Motion Detector, Temperature Monitor, Smoke Detector, Smart Light, Smart Fan, Smart Air Conditioner, Smart Door and Webcam.

Automation Rules:

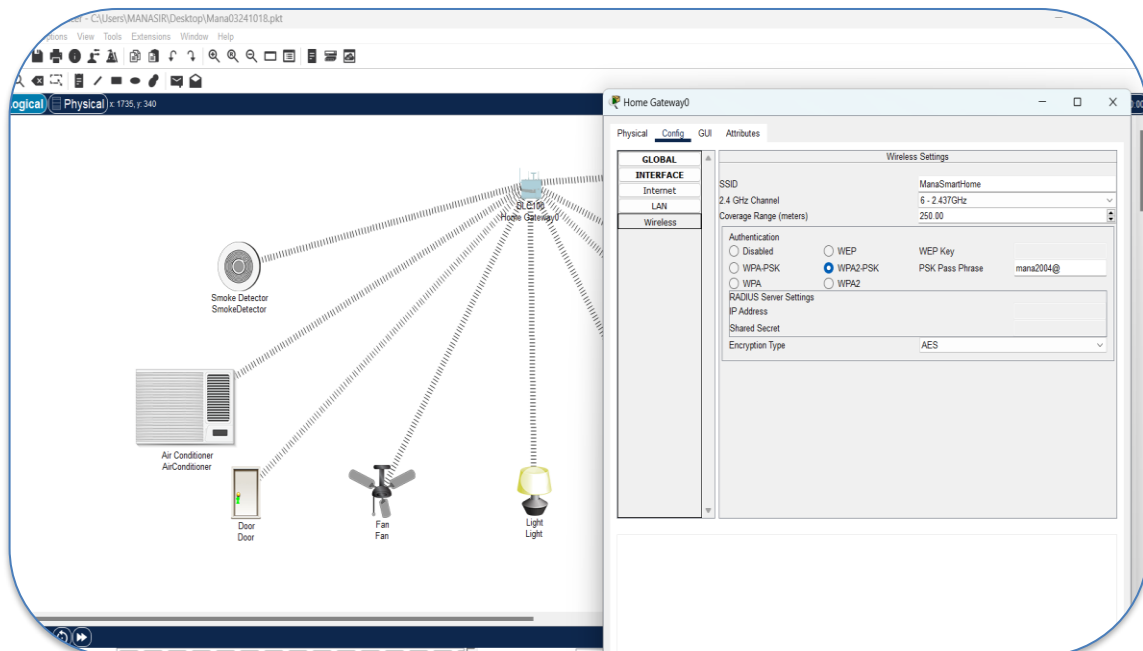
1. Motion detected -> Light ON and Webcam ON.
2. Temperature above 30°C -> Fan ON and Air Conditioner ON.
3. Smoke detected -> Door OPEN and Webcam ON.

Devices communicate through the Home Gateway, allowing centralized monitoring and automation.

3.1 – Complete Smart Home IoT Network



3.2 – Home Gateway Configuration



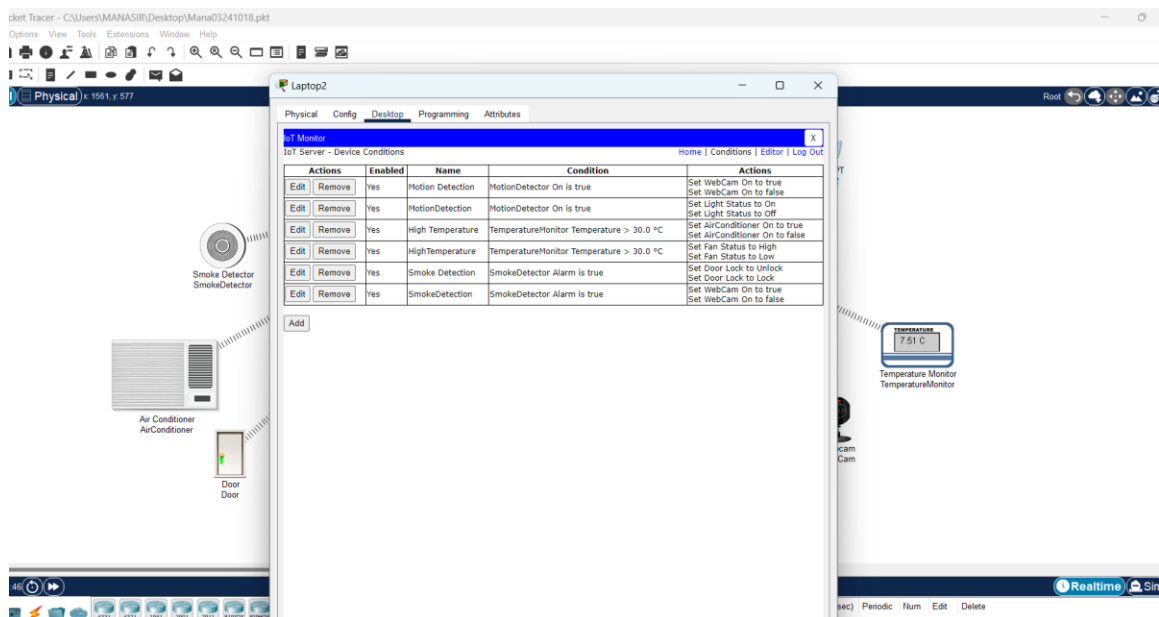
3.3 – Automation Rules/Conditions

The Smart Home system uses automation rules to improve security, comfort, and energy efficiency. These rules allow IoT devices to respond automatically based on sensor readings without requiring manual intervention. The Home Gateway continuously monitors the sensors and executes the predefined actions whenever the specified conditions are met.

The implemented automation rules are:

- Rule 1: When the Motion Detector detects movement, the Smart Light and Webcam are automatically turned ON. This provides better visibility and enhances home security.
- Rule 2: When the Temperature Monitor detects a temperature above 30°C, the Smart Fan and Smart Air Conditioner are automatically switched ON to maintain a comfortable indoor environment.
- Rule 3: When the Smoke Detector detects smoke, the Smart Door automatically opens and the Webcam is activated. This supports emergency evacuation and enables real-time monitoring during fire-related incidents.

These automation rules demonstrate how IoT technology can make a home more intelligent by reducing manual operations, improving safety, and increasing energy efficiency.



Chapter 4 – VR Simulation

Environment Design:

A virtual smart home was created using Delightex EDU. Users can navigate through the house and interact with smart devices.

Interactive Elements:

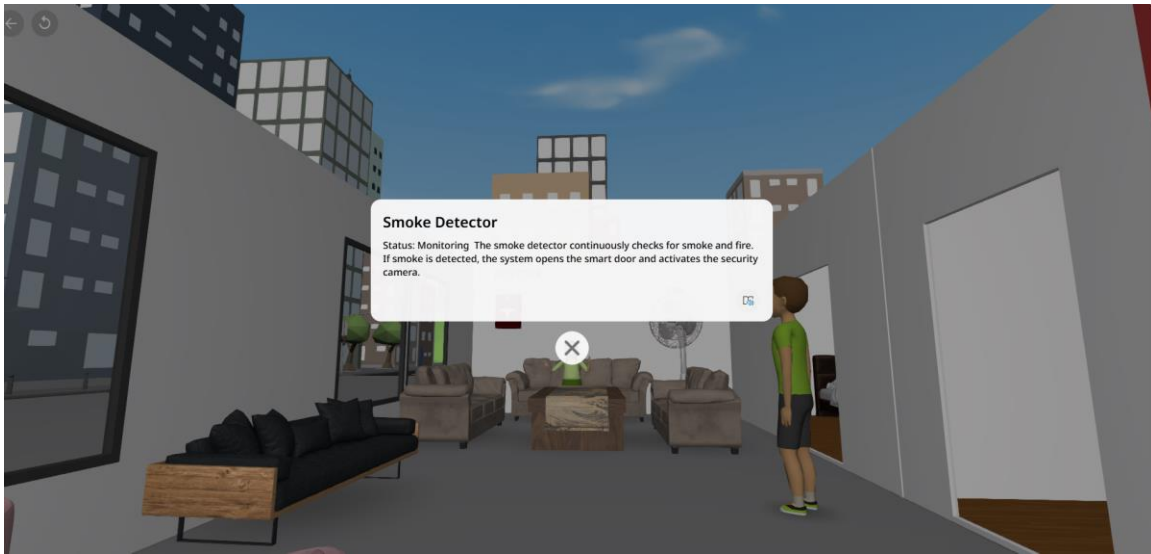
- Smart Door
- Motion Detector
- Smoke Detector
- Temperature Sensor
- Fan
- Air Conditioner
- Webcam
- Laptop Dashboard

Each object displays information explaining its role in the smart home.

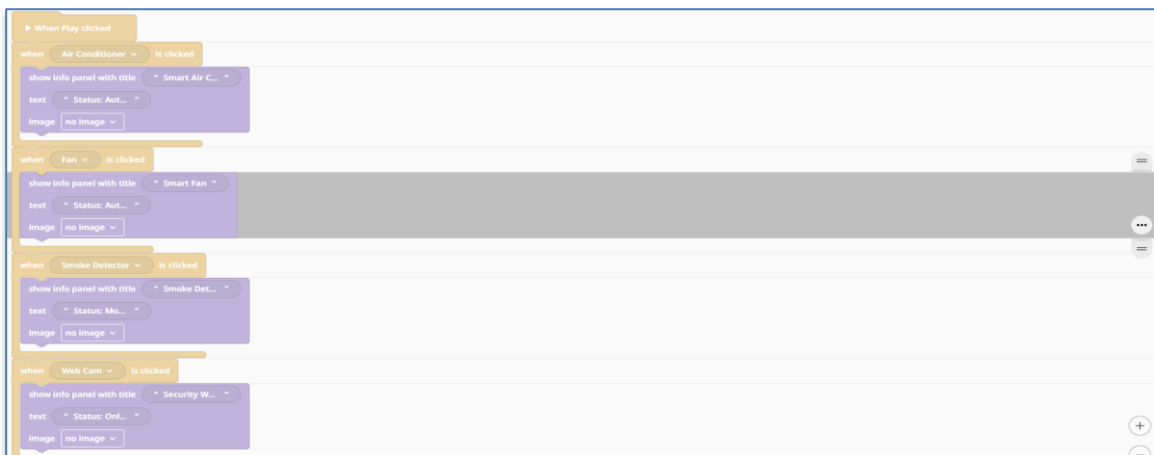
4.1 – Virtual Smart Home Environment

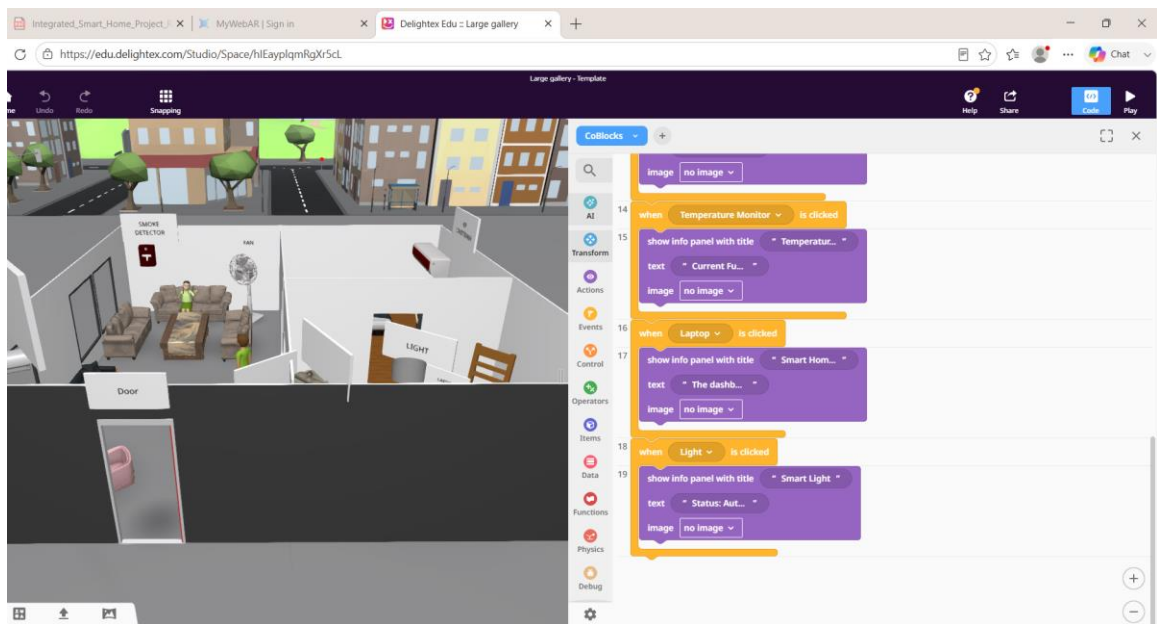
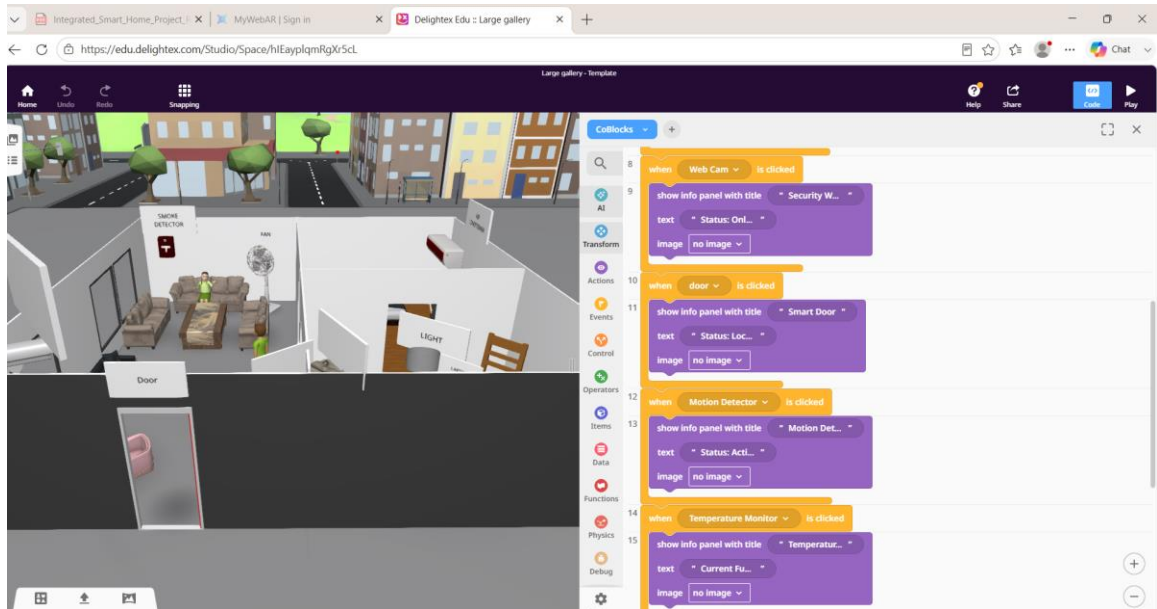


4.2 – Interactive Device



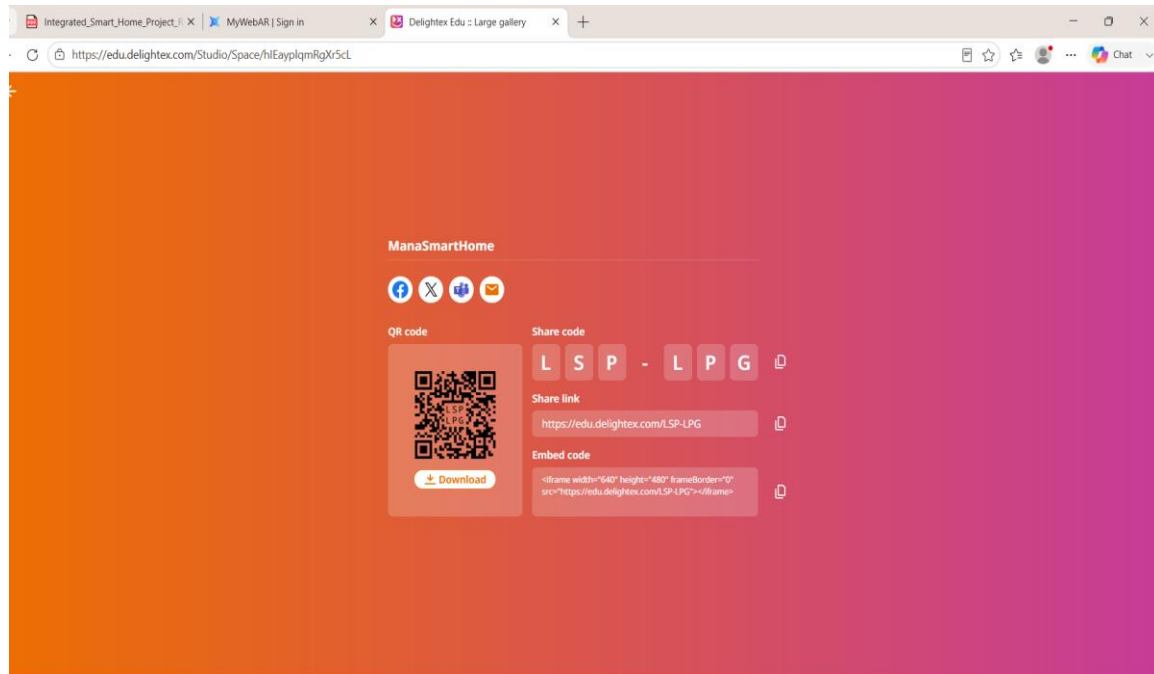
4.2- CoBlocks Configuration for Interactive Smart Home Device





4.3- Delightex EDU Project Link

VR Project Link - <https://edu.delightex.com/LSP-LPG>



The link above opens the live, editable Delightex EDU workspace for this project and can be used to re-enter the VR walkthrough for demonstration or further editing

Chapter 5 – AR Simulation

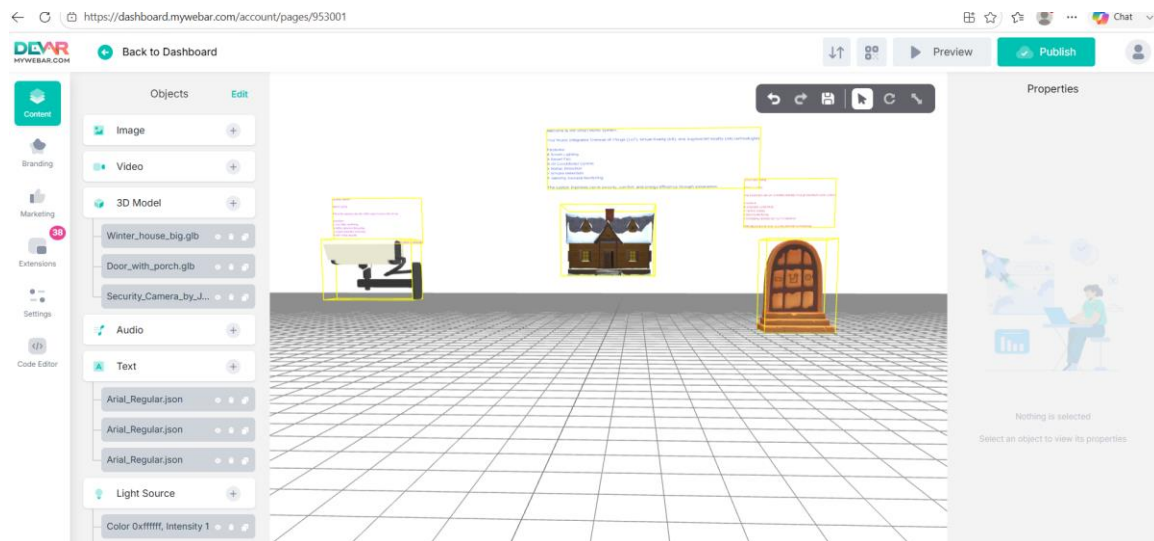
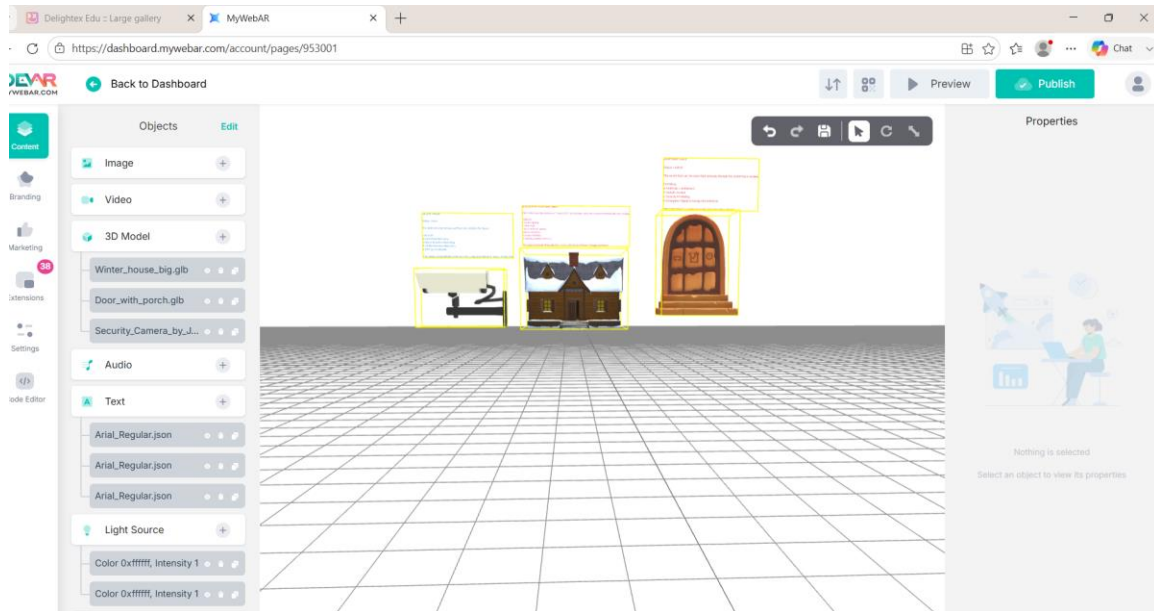
AR Design:

MyWebAR was used to create an augmented reality scene containing a Smart Home model, Smart Door and Security Camera.

User Interaction:

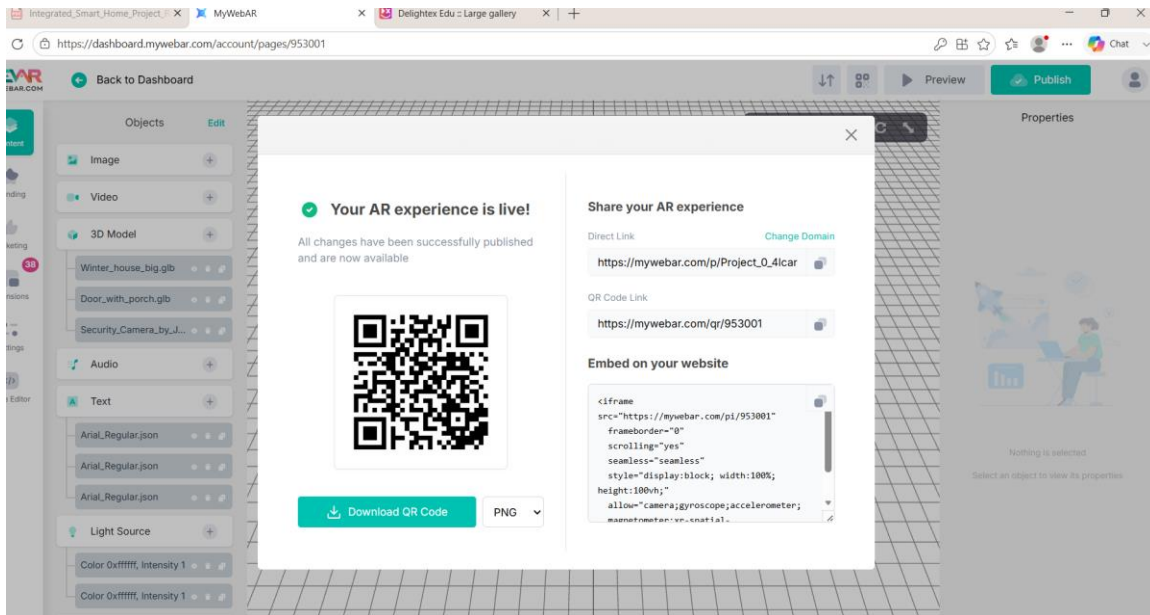
Users can tap each object to view descriptions and operational information, improving understanding of the system.

5.1 – Smart Home AR Scene



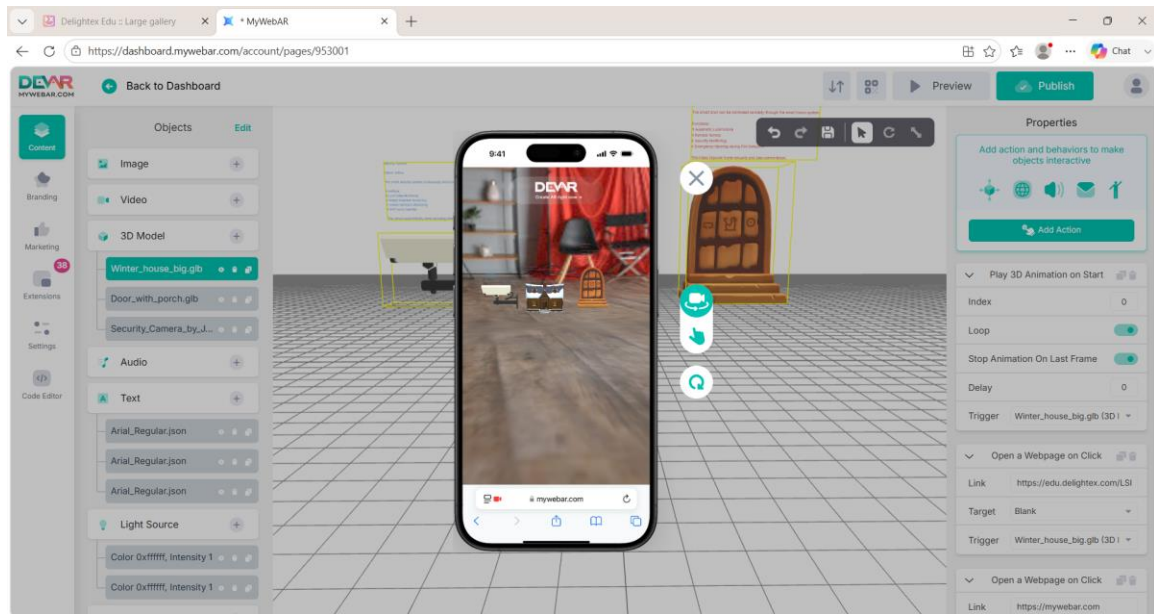
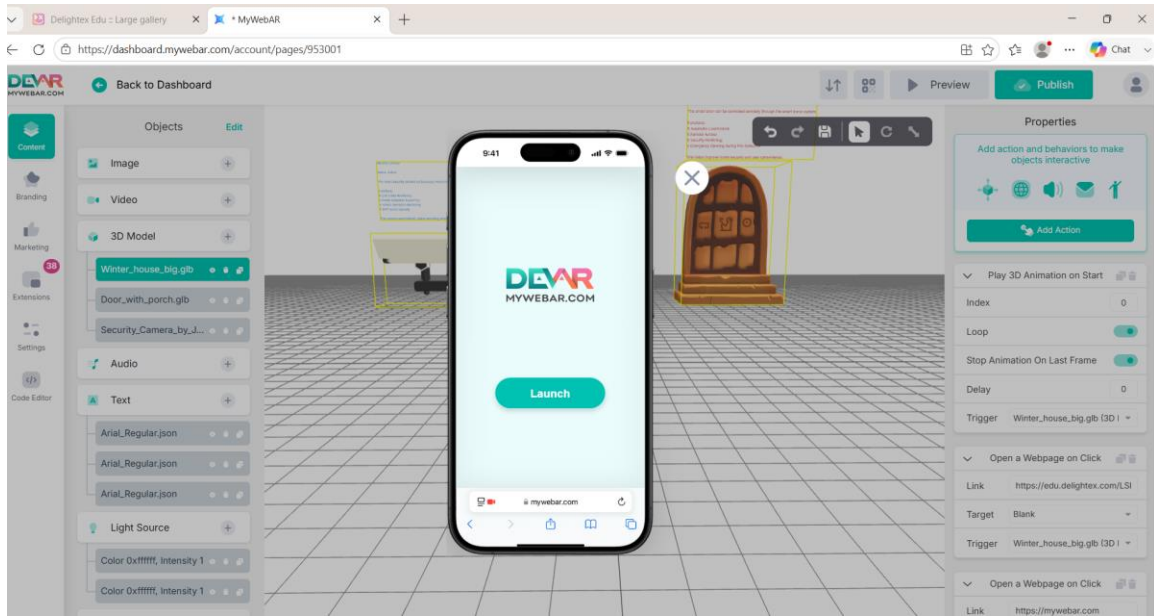
5.2 Information Overlay & Publishing

Once finalised, the experience was published directly from MyWebAR, generating a shareable link and a scannable QR code so the AR experience can be opened on any mobile device without an app install.



5.3 User Interaction

The AR scene was tested live through a mobile browser: pointing the camera at the target reveals the 3D device models in the real-world camera feed along with their descriptive overlay text, for example the Smart Web Camera and the overall Smart Home description.



Chapter 6 – Integration

The IoT system collects sensor data and controls smart devices. The VR environment provides a virtual walkthrough of the home, while the AR application overlays information about smart devices. Together these technologies improve automation, visualization, security, maintenance and user experience.

Benefits:

- Improved security
- Energy efficiency
- Remote monitoring
- Interactive visualization
- Better user awareness

Chapter 7 – Conclusion

This project successfully demonstrates how IoT, Virtual Reality and Augmented Reality can be integrated into a Smart Home solution. Automation improves comfort and security while VR and AR enhance visualization and interaction.

Challenges:

- Learning multiple platforms
- Device configuration
- Integration planning

Future Improvements:

- Mobile application integration
- Voice assistant support
- Cloud storage
- AI-based predictive automation

References

Cisco Networking Academy. Cisco Packet Tracer.
Delightex EDU Documentation.

MyWebAR Documentation.
IEEE Internet of Things resources.